

AMENDMENT No. 3 (06/2026)

To

AIS-128 Testing Procedure and Requirements for Headlamp Beam Testing and Brake Testing at Automated Testing Stations using Headlight Tester and Roller Brake Tester

1.0 Page 1/5, Heading:

Substitute following text for existing text

Testing Procedure and Requirements for Headlamp Beam Testing and Brake Testing at Automated Testing Stations using Headlight Tester and Roller Brake Tester

2.0 Page 3/5, Clause 3.4: Pass/ Fail Criteria

Substitute following text for existing text

“Service brake efficiency when measured on roller brake tester should be more than 40%.”

3.0 Page 4/5 Addition of two wheeled vehicles brake testing procedure as Part-1 in AIS-128

AIS-128 (Part-1) Testing Procedure and Requirements for Brake Testing of Two Wheeled Vehicles at Automated Testing Stations using Roller Brake Tester

1.0 Scope:

This standard lays down the testing procedure and requirements for braking testing for fitness testing of two-wheeled vehicles of category L1 and L2 as defined in CMVR at Automated Testing Stations using roller brake tester.

2.0 Test Requirements:

2.1 This inspection shall be carried out by means of a roller brake tester.

2.2 When using the brake tester for this inspection, the following aspects should be considered

2.1.1 The braking test shall not be carried out when any of the tires shows a degree of deformation due to visually appreciable deflation.

2.1.2 The test shall be carried out by operating the necessary controls to exert the braking action on all braked wheels independently if they cannot both fit in the roller brake tester. In these cases, the efficiency calculation is the total from both wheels when operated by that control only.

2.1.3 Incorrect tire pressure can give rise to erroneous readings, so correct tire pressure is necessary. The tread must also have a tread pattern.

2.1.4 In some cases of vehicles equipped with an anti-lock braking system, the system malfunction lamp may come on when the brake rollers of the brake tester come into operation. To correct this problem, once the vehicle has exited the brake tester, the engine should be stopped and the engine restarted,

whereupon the warning lamp should go out after the system check. In some cases, it might take a short run for the warning lamp to go out.

3.0 **Test Procedure:**

- i) Position the front wheel of the vehicle on the roller brake tester equipment.
- ii) Shift the vehicle to neutral and then release the vehicle brake pedal / lever.
- iii) Start the engine and keep the engine running at idling.
- iv) The rollers will rotate the wheels at the pre-set speed of min. 5 km/h.
- v) Apply the brake firmly to stop the brake tester rollers.
- vi) The maximum brake force required to stop front wheel of the vehicle shall be recorded by the roller brake tester.
- vii) Release the brake and drive the vehicle forward until the rear wheel is on the rollers.
- viii) Repeat the same steps from ii) to vii) for rear wheel also.
- ix) Total braking efficiency of a vehicle is calculated from the sum of all braking forces of both the wheels and total weight of vehicle measured on roller brake tester.
- x) Drive the vehicle off the roller brake tester.

4.0 **Braking Efficiency:**

Braking Efficiency (E) means the ratio of the braking forces to the weight of vehicle measured on roller brake tester.

$$E = \left(\frac{F}{W} \right) * 100$$

E = Braking efficiency in %

F = Sum of all braking forces in Newtons (sum of brake tester readings for both wheels in N)

W = Weight of vehicle measured on roller brake tester in N

5.0 **Pass / Fail Criteria:**

Total Braking efficiency of two-wheeled vehicles when measured on a roller brake tester should be more than 40%.

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P. B. NO. 832, PUNE 411 004

ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

June 2026

AMENDMENT NO. 2 (07/2023)

TO

**AIS-128 : Testing Procedure and Requirements for Headlamp Beam Testing and
Brake Testing at Authorized Testing Stations using Headlight Tester and
Roller Brake Tester**

1. Page 1/5, Clause 2.0, Title:

Delete “Manual” word from the existing text, as follows:

**“HEADLAMP BEAM TESTING PROCEDURE (USING SEMI
AUTOMATIC / AUTOMATIC HEADLIGHT TESTER)”**

2. Page 1/5, Clause 2.1:

Delete “Manual” word from the existing text, as follows:

“2.1 The procedure below describes the testing of headlamp beam using headlight
tester (semi-automatic / automatic).

3. Page 1/5, Clause 2.2:

Delete “Manual” word from the existing text, as follows:

“2.2 Test Overview

The headlight tester (semi-automatic / automatic) is capable of measuring the
horizontal cut-off of the passing beam (low beam) when measured at a distance
specified by instrument manufacturer. It shall simulate the condition of testing
of passing beam when tested at 10m as specified in Annexure VII of CMVR.
Brief description of headlight tester is given in clause 4.1.”

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING
COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

13th July 2023

Amendment No. 1 5 June 2015

TO

AIS -128

**Testing Procedure and Requirements for Headlamp Beam Testing and
Brake Testing at Authorized Testing Stations using Headlight Tester
and Roller Brake Tester**

1.0 Page 2/5, clause 3.2:

Substitute following text for the exiting text:

“ 3.2 Test Overview

Roller brake tester measures the braking force required to stop the vehicle. Braking efficiency shall be established for service and parking brakes. Roller brake tester shall also measure the brake imbalance. Brief description of roller brake tester is given in clause 4.2. The test shall be carried out with vehicle in unladen condition.”

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UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
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5 June 2015

AUTOMOTIVE INDUSTRY STANDARD

**Testing Procedure and Requirements for
Headlamp Beam Testing and Brake Testing
at Authorized Testing Stations using
Headlight Tester and Roller Brake Tester**

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

February 2014

Status chart of the standard to be used by the purchaser for updating the record

Sr. No.	Corrigenda.	Amendment	Revision	Date	Remark	Misc.
General remarks :						

INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of standards and development of test facilities in parallel when the work on the preparation of the standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MoST) has constituted a permanent Automotive Industry Standards Committee (AISC) vide order No. RT-11028/11/97-MVL dated September 15, 1997. The standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CTSC). After approval, the Automotive Research Association of India (ARAI), Pune, being the secretariat of the AIS Committee, has published this standard. For better dissemination of this information ARAI has published this document on their Web site.

Ministry of Road Transport & Highways (MoRTH), Government of India has initiated establishment of 10 automated Inspection & Certification (I&C) Test Centers where verification of vehicle fitness is carried out using test lane equipment. In these centers, several instruments are used to verify the road worthiness of in-use vehicles. The tests carried out at minimum are as per regulations defined in CMVR 62. Such centers will be useful to introduce further rules for improving our road worthiness requirements. The subject of formulation of test procedures and requirements for vehicle systems in Inspection & Certification (I&C) Test Centers was discussed in CMVR-TSC and AISC. Accordingly this standard is formulated to cover test procedures and requirements for headlamp beam testing and brake testing at authorized testing station using headlight tester and roller brake tester.

The Automotive Industry Standards Committee (AISC) responsible for preparation of this standard are given in Annex I.

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**Testing Procedure and Requirements for Headlamp Beam
Testing and Brake Testing at Authorized Testing Stations
using Headlight Tester and Roller Brake Tester**

1.0 SCOPE

This standard lays down the testing procedure and requirements for headlamp beam testing and brake testing for fitness check of vehicles at authorized testing stations using headlight tester and roller brake tester.

2.0 HEADLAMP BEAM TESTING PROCEDURE (USING MANUAL / SEMI-AUTOMATIC / AUTOMATIC HEADLIGHT TESTER)

2.1 The procedure below describes the testing of headlamp beam using headlight tester (manual / semi-automatic / automatic).

2.2 Test Overview

The headlight tester (manual / semi-automatic / automatic) is capable of measuring the horizontal cut-off of the passing beam (low beam) when measured at a distance specified by instrument manufacturer. It shall simulate the condition of testing of passing beam when tested at 10m as specified in Annexure VII of CMVR. Brief description of headlight tester is given in clause 4.1.

2.3 Test Procedure

- i) Position the vehicle at a distance as specified in clause 2.2 from headlight tester and perpendicular to the headlight tester track.
- ii) Turn on the headlamp low beam.
- iii) The headlight tester is to be positioned in front of the headlamp position and begin testing of headlamp.
- iv) The headlight tester will measure the headlamp's low beam pattern with respect to center line in terms of a unit termed % deviation.
- v) The headlight tester is now to be positioned in front of the other headlamp, if any.
- vi) The headlight tester then tests the headlamp's low beam on other side of the vehicle repeating steps iii) and iv).
- vii) The recorded test data will be stored.
- viii) After the test gets completed, the headlight tester will be moved to its rest position.
- ix) The vehicle shall be moved to next station.

2.4 Passing Criteria

Horizontal cut off of passing beam shall always be below headlamp centerline and the deviation shall be within 0.5% to 2.5%.

3.0 BRAKE TESTING USING ROLLER BRAKE TESTER

3.1 The procedure below describes the process of conducting the vehicle brake testing using a roller brake tester.

3.2 Test Overview

Roller brake tester measures the braking force required to stop the vehicle. Braking efficiency shall be established for service and parking brakes. Roller brake tester shall also measure the brake imbalance. Brief description of roller brake tester is given in clause 4.2.

3.3 Test Procedure

- i) Position the front axle of the vehicle on the roller brake tester
- ii) Shift the vehicle to neutral and then release the vehicle brake pedal.
- iii) Start the engine and keep the engine running at idling.
- iv) In the case of power assisted brakes (air brakes, air assisted, vacuum assisted etc.) press the accelerator pedal and keep the engine running till the air/vacuum reservoirs are charged fully.
- v) Run both sets of rollers together in a forward direction to align the vehicle
- vi) Then the rollers will rotate wheels at the preset speed (approx. 5 km/h for 3 and 4 wheeled vehicles having GVW not exceeding 3500 kg and approx. 2.5 km/h for 4 wheeled vehicles having GVW exceeding 3500 kg).
- vii) Apply the service brake firmly to stop the brake tester rollers.
- viii) The maximum brake force required to stop each wheel on the axle shall be recorded by the roller brake tester.
- ix) Release the service brake and drive the vehicle forward until the rear axle wheels are on the rollers.
- x) Repeat the same steps from ii) to viii) for rear axle also.
- xi) Release the service brake.
- xii) Total braking efficiency of vehicle is calculated from the sum of all braking forces of all wheels and total weight of vehicle. Also the difference between the braking efficiency of two wheels on the same axle will be calculated.
- xiii) As the parking brake is installed on the rear axle itself, repeat the steps from ii) to viii) for calculating the braking efficiency of the parking brake.
- xiv) Release the parking brake.
- xv) Drive the vehicle off the roller brake tester.

The vehicle shall be considered as meeting the requirement only if the total braking efficiency of service brake is as per the pre-defined values.

3.4 **Pass/ Fail Criteria**

Breaking efficiency when measured on roller brake tester should be more than 27.23%.

4.0 **BRIEF DRESCRIPTION AND FEATURES OF HEADLIGHT TESTER AND ROLLER BRAKE TESTER**

4.1 **Requirements for Headlight Tester**

4.1.1 **Brief description**

Headlamp testing apparatus shall consist mainly of a collimating lens and a marked screen. With this system it is possible to project the light distribution (light/dark limit) from the headlamps and to check the inclination of the beam.

The apparatus shall be capable of vertical adjustment to test headlamps of various heights. The instrument output shall be in terms of percentage inclination of headlamp beam.

4.1.2 **Required features**

- i) Height: The adjustment range of the headlight tester must be capable of measuring headlamps with their centres 500 mm to 1200 mm above ground level.
- ii) Inclination (vertical orientation) maximum deviation: The maximum deviation of inclination for vertical orientation shall not be more than $\pm 0.1\%$.
- iii) Alignment with the longitudinal axis of the vehicle: The apparatus shall be capable of being accurately aligned with the longitudinal axis of the vehicle.

4.2 **Requirements for Roller Brake Testers**

4.2.1 **Brief description**

Roller brake testers are required on test lanes where Light and Heavy Vehicles are inspected. Roller brake testing machine consist of two pairs of rollers on which the wheels of an axle are placed during test. The braking force produced by the wheels is converted directly into an indication of braking efficiency.

4.2.2 **Required features**

- i) Set of rollers of adequate diameter with suitable surface finish, connected via suitable drive configuration to a motor of adequate capacity in order to carry out testing at desired testing speed

- ii) The braking torque applied by the motor measured by suitable sensor at point of slipping.
- iii) A suitable arrangement may be provided for positioning the vehicle on the test rig and taking it off the test rig after the testing is over.
- iv) Measurement Display: The console should display individual brake force for each of the wheels on the axle and braking efficiency. The display shall be readily visible to the inspector when seated in the driving seat of the vehicle being inspected.

ANNEX I

(See introduction)

COMMITTEE COMPOSITION *
Automotive Industry Standards Committee

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Shri Shrikant R. Marathe	Director, The Automotive Research Association of India, Pune
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* At the time of approval of this Automotive Industry Standard (AIS)